

book_assistant_project

December 15, 2025

1 Book Recommender Assistant Project

1.1 00 - Environment Setup

1.1.1 00.1 - Install Requirements

```
[ ]: import sys, textwrap, subprocess, pkg_resources

print("Using Python:", sys.executable)
print("Python version:", sys.version)

# --- Core list of required packages ---
REQUIRED_PACKAGES = [
    # Core numerics / data
    "torch>=2.2",
    "numpy>=1.26",
    "pandas>=2.2",
    "pyarrow>=15.0",
    "fastparquet",

    # Embeddings + transformers
    "sentence-transformers>=3.0.1",
    "transformers>=4.45.0",
    "tokenizers>=0.21,<0.22",

    # ML utilities
    "scikit-learn>=1.4",
    "tqdm>=4.65",
    "requests>=2.32",
    "huggingface-hub>=0.23",

    # Notebook / misc
    "ipykernel",
    "ipython",
    "jupyter",
    "rich>=13.0",
]
```

```

print("\nInstalling / upgrading packages (this may take a few minutes)...\n")

cmd = [sys.executable, "-m", "pip", "install", "-U"] + REQUIRED_PACKAGES
print(" ".join(cmd))
subprocess.check_call(cmd)

print("\n pip install finished.")

# Show versions of a few key libs
def show_version(pkg_name):
    try:
        v = pkg_resources.get_distribution(pkg_name).version
        print(f" {pkg_name}: {v}")
    except Exception:
        print(f" {pkg_name}: NOT FOUND")

print("\nInstalled versions:")
for name in ["torch", "numpy", "pandas", "sentence-transformers",
            ↪ "transformers", "tokenizers"]:
    show_version(name)

print("\n When this finishes, it's a good idea to RESTART the kernel before ↪
↪ continuing.")

```

1.1.2 00.2 - Check Imports

```

[13]: try:
        import torch
        import numpy as np
        import pandas as pd
        from sentence_transformers import SentenceTransformer
        from transformers import AutoModel, AutoTokenizer
        print(" Core ML imports OK")
    except Exception as e:
        print(" Problem importing ML libs:", repr(e))

    try:
        import build_desc_embeddings
        print(" build_desc_embeddings imported")
    except Exception as e:
        print(" Could not import build_desc_embeddings:", repr(e))

    try:
        from pseudo_user_teacher import DescEmbeddingTeacher
        print(" pseudo_user_teacher.DescEmbeddingTeacher imported")
    except Exception as e:
        print(" Could not import DescEmbeddingTeacher:", repr(e))

```

```

try:
    from overarching_module import BookRecSession
    print(" overarching_module.BookRecSession imported")
except Exception as e:
    print(" Could not import BookRecSession:", repr(e))

```

```

Core ML imports OK
build_desc_embeddings imported
pseudo_user_teacher.DescEmbeddingTeacher imported
overarching_module.BookRecSession imported

```

1.1.3 00.3 - Configure Ollama

```

[ ]: import shutil, sys, subprocess

def run_cmd(cmd):
    print("\n$", " ".join(cmd))
    try:
        out = subprocess.check_output(cmd, stderr=subprocess.STDOUT, text=True)
        print(out)
    except subprocess.CalledProcessError as e:
        print(e.output)

if shutil.which("ollama") is None:
    print(" 'ollama' command not found on PATH.")
    print("   Install from https://ollama.com/ and ensure `ollama` is callable_
↳in this environment.")
else:
    print(" Ollama found on PATH.")

    # Show version
    run_cmd(["ollama", "--version"])

    # Pull the models you use in planner / RAG
    # Adjust this list if your planner uses different models.
    MODELS = [
        "mistral",          # planner LLM
        "nomic-embed-text", # embedding model (if used)
    ]

    for m in MODELS:
        print(f"\n=== Ensuring Ollama model '{m}' is available ===")
        run_cmd(["ollama", "pull", m])

print("\n   Make sure `ollama serve` is running in the background when you use_
↳the planner.")

```

1.2 01 - Data Preparation for the Book Assistant

```
[1]: from pathlib import Path
import sys
import importlib.util

# Assume notebook is opened from the project root
PROJECT_ROOT = Path.cwd().resolve()

print("Project root:", PROJECT_ROOT)

DATA_DIR = PROJECT_ROOT / "data"
RAW_DIR = DATA_DIR / "raw"
PROC_DIR = DATA_DIR / "processed"

print("DATA_DIR:", DATA_DIR)
print("RAW_DIR:", RAW_DIR)
print("PROC_DIR:", PROC_DIR)

# Make sure these exist
RAW_DIR.mkdir(parents=True, exist_ok=True)
PROC_DIR.mkdir(parents=True, exist_ok=True)

# Show python executable being used
print("Python executable:", sys.executable)

def load_module_from_path(path: Path, module_name: str):
    """Dynamically import a module from a given path."""
    path = Path(path)
    if not path.exists():
        raise FileNotFoundError(f"Module file not found: {path}")

    spec = importlib.util.spec_from_file_location(module_name, str(path))
    module = importlib.util.module_from_spec(spec)
    assert spec.loader is not None
    spec.loader.exec_module(module)
    return module

# Load data_prep.py from data/
data_prep_path = DATA_DIR / "data_prep.py"
data_prep = load_module_from_path(data_prep_path, module_name="data_prep")

print("Loaded data_prep from:", data_prep_path)
print("data_prep has:", [name for name in dir(data_prep) if name.
    ↳startswith("run_") or name.endswith("_goodreads")])
```

Project root: /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM

```

DATA_DIR: /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project
/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data
RAW_DIR: /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/
Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/raw
PROC_DIR: /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project
/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/processed
Python executable: /usr/local/bin/python3
Loaded data_prep from: /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/data_prep.py
data_prep has: ['download_raw_goodreads', 'run_pipeline']

```

```

[11]: # Toggle this flag depending on whether you want to auto-download from UCSD
      DOWNLOAD_RAW = True # set to False if you've already downloaded data

```

```

data_prep.download_raw_goodreads(download=DOWNLOAD_RAW)

```

```

[download] some raw Goodreads files missing; downloading from UCSD...
[download] already exists: /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/raw/goodreads_books.json.gz
[download] downloading goodreads_book_authors.json.gz from https://mcauleylab.ucsd.edu/public_datasets/gdrive/goodreads/goodreads_book_authors.json.gz
[download] goodreads_book_authors.json.gz: 100.00%
[download] already exists: /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/raw/goodreads_interactions_dedup.json.gz

```

```

[12]: import subprocess

prep_all_path = DATA_DIR / "prep_all.py"

if not prep_all_path.exists():
    raise FileNotFoundError(f"prep_all.py not found at {prep_all_path}")

cmd = [sys.executable, str(prepare_all_path)]
print("Running:", " ".join(cmd))

result = subprocess.run(cmd)
if result.returncode != 0:
    raise RuntimeError(f"prep_all.py failed with exit code {result.returncode}")
else:
    print("prep_all.py finished successfully.")

```

```

Running: /usr/local/bin/python3 /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/prep_all.py

```

```

=== [STEP] Base data prep (books + interactions) ===

```

```

/usr/local/bin/python3 /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/data_prep.py
[download] download=False; skipping raw downloads.
=== data_prep.py: base data preparation ===
[books] loading from /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/raw/goodreads_books.json.gz
[books] streaming JSON from /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/raw/goodreads_books.json.gz
[books] first chunk shape: (100000, 29)
[books] using language column: language_code
[books] using popularity column: ratings_count
[books] processed 1000000 rows, current top_df size=366607
[books] processed 2000000 rows, current top_df size=309895
[books] selected top 200000 English books (mean rating=3.940, median ratings_count=575.0)
   book_id  ... authors
0  2767052  ... [{"author_id": "153394", "role": ""}]
1         3  ... [{"author_id": "1077326", "role": ""}, {"author_...
2   41865  ... [{"author_id": "941441", "role": ""}]
3    2657  ... [{"author_id": "1825", "role": ""}]
4    4671  ... [{"author_id": "3190", "role": ""}]
5 11870085  ... [{"author_id": "1406384", "role": ""}]
6    5907  ... [{"author_id": "656983", "role": ""}]
7    5107  ... [{"author_id": "819789", "role": ""}]
8    1885  ... [{"author_id": "1265", "role": ""}, {"author_i...
9     960  ... [{"author_id": "630", "role": ""}]

[10 rows x 6 columns]
[interactions] loading from /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/raw/goodreads_interactions_dedup.json.gz
[interactions] streaming JSON from /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/raw/goodreads_interactions_dedup.json.gz
[interactions] reached cap of 500000 kept rows; stopping early.
[interactions] final (typed + filtered) shape: (500000, 11)
[interactions] unique users: 2582
[interactions] unique books: 98949
[save] books -> /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/processed/books.parquet
[save] interactions -> /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/processed/interactions.parquet
[done] data_prep.py complete.

```

[OK] Base data prep (books + interactions) finished successfully.

=== [STEP] Build authors.parquet ===

/usr/local/bin/python3 /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/build_authors.py

[load] Loading authors from /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/raw/goodreads_book_authors.json.gz

[info] raw authors columns: ['average_rating', 'author_id', 'text_reviews_count', 'name', 'ratings_count']

[info] authors deduped on author_id: 829529 -> 829529

[done] Saved authors parquet to /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/processed/authors.parquet

	author_id	name
0	604031	Ronald J. Fields
1	626222	Anita Diamant
2	10333	Barbara Hambly
3	9212	Jennifer Weiner
4	149918	Nigel Pennick

[OK] Build authors.parquet finished successfully.

=== [STEP] Enrich books_with_genres.parquet ===

/usr/local/bin/python3 /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/enrich_books_genres.py

[load] Loading books...

[compute] Assigning real or synthetic genres...

[info] Sample rows with genres:

	book_id	...	genres
0	2767052	...	[fantasy, science fiction, romance, mystery, y...
1	3	...	[fantasy, science fiction, mystery, young adul...
2	41865	...	[fantasy, science fiction, romance, young adul...
3	2657	...	[mystery, young adult, nonfiction, historical,...
4	4671	...	[romance, young adult, historical, classics]
5	11870085	...	[romance, young adult]
6	5907	...	[fantasy, science fiction, young adult, nonfic...
7	5107	...	[young adult, historical, classics]
8	1885	...	[romance, young adult, nonfiction, historical,...
9	960	...	[fantasy, science fiction, romance, mystery, n...

[10 rows x 4 columns]

[save] Saving enriched books to /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/processed/books_with_genres.parquet

[done] books_with_genres.parquet updated with real+synthetic genres.

[OK] Enrich books_with_genres.parquet finished successfully.

All data prep steps completed. You should now have:

- data/processed/books.parquet (or similar)
- data/processed/interactions.parquet
- data/processed/authors.parquet
- data/processed/books_with_genres.parquet

prep_all.py finished successfully.

1.3 02 - Generate Document Embeddings For Book Similarity

```
[ ]: # import sys

# # Upgrade transformers + sentence-transformers together
# !{sys.executable} -m pip install -U "transformers>=4.45.0"
# ↪ "sentence-transformers>=3.0.0"
```

Requirement already satisfied: transformers>=4.45.0 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(4.57.3)

Requirement already satisfied: sentence-transformers>=3.0.0 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(5.2.0)

Requirement already satisfied: filelock in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from transformers>=4.45.0) (3.16.1)

Requirement already satisfied: huggingface-hub<1.0,>=0.34.0 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from transformers>=4.45.0) (0.36.0)

Requirement already satisfied: numpy>=1.17 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from transformers>=4.45.0) (1.26.4)

Requirement already satisfied: packaging>=20.0 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from transformers>=4.45.0) (24.1)

Requirement already satisfied: pyyaml>=5.1 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from transformers>=4.45.0) (6.0.2)

Requirement already satisfied: regex!=2019.12.17 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from transformers>=4.45.0) (2024.9.11)

Requirement already satisfied: requests in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from transformers>=4.45.0) (2.32.3)

Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from transformers>=4.45.0) (0.22.1)

Requirement already satisfied: safetensors>=0.4.3 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages

(from transformers>=4.45.0) (0.5.3)
Requirement already satisfied: tqdm>=4.27 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from transformers>=4.45.0) (4.66.5)
Requirement already satisfied: fsspec>=2023.5.0 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from huggingface-hub<1.0,>=0.34.0->transformers>=4.45.0) (2024.9.0)
Requirement already satisfied: typing-extensions>=3.7.4.3 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from huggingface-hub<1.0,>=0.34.0->transformers>=4.45.0) (4.15.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from huggingface-hub<1.0,>=0.34.0->transformers>=4.45.0) (1.2.0)
Requirement already satisfied: torch>=1.11.0 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from sentence-transformers>=3.0.0) (2.6.0)
Requirement already satisfied: scikit-learn in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from sentence-transformers>=3.0.0) (1.7.2)
Requirement already satisfied: scipy in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from sentence-transformers>=3.0.0) (1.13.1)
Requirement already satisfied: networkx in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from torch>=1.11.0->sentence-transformers>=3.0.0) (3.3)
Requirement already satisfied: jinja2 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from torch>=1.11.0->sentence-transformers>=3.0.0) (3.1.4)
Requirement already satisfied: setuptools in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from torch>=1.11.0->sentence-transformers>=3.0.0) (75.1.0)
Requirement already satisfied: sympy==1.13.1 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from torch>=1.11.0->sentence-transformers>=3.0.0) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from sympy==1.13.1->torch>=1.11.0->sentence-transformers>=3.0.0) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from jinja2->torch>=1.11.0->sentence-transformers>=3.0.0) (2.1.5)
Requirement already satisfied: charset-normalizer<4,>=2 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from requests->transformers>=4.45.0) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from requests->transformers>=4.45.0) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages

```
(from requests->transformers>=4.45.0) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from requests->transformers>=4.45.0) (2024.8.30)
Requirement already satisfied: joblib>=1.2.0 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from scikit-learn->sentence-transformers>=3.0.0) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages
(from scikit-learn->sentence-transformers>=3.0.0) (3.5.0)
```

```
[4]: from pathlib import Path
import importlib.util

# Assuming you already computed PROJECT_ROOT earlier
BUILD_DESC_PATH = PROJECT_ROOT / "build_desc_embeddings.py"
print("build_desc_embeddings.py path:", BUILD_DESC_PATH)

def load_module_from_path(path: Path, module_name: str):
    path = Path(path)
    if not path.exists():
        raise FileNotFoundError(f"Module file not found: {path}")
    spec = importlib.util.spec_from_file_location(module_name, str(path))
    module = importlib.util.module_from_spec(spec)
    assert spec.loader is not None
    spec.loader.exec_module(module)
    return module

build_desc = load_module_from_path(BUILD_DESC_PATH, "build_desc_embeddings")

print("Loaded build_desc_embeddings. Public API:")
[name for name in dir(build_desc) if name.startswith("build_")]
```

```
build_desc_embeddings.py path: /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE25
9_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/
build_desc_embeddings.py
Loaded build_desc_embeddings. Public API:
```

```
[4]: ['build_description_embeddings', 'build_text_corpus']
```

```
[5]: import pandas as pd
import numpy as np

embs = build_desc.build_description_embeddings(limit_books=None)
print("Embeddings shape:", embs.shape)

books_path = PROJECT_ROOT / "data" / "processed" / "books_with_genres.parquet"
emb_path = PROJECT_ROOT / "models" / "desc_embeddings.npy"
```

```
df_books = pd.read_parquet(books_path)
embs = np.load(emb_path)

print("books_with_genres shape:", df_books.shape)
print("embeddings shape:", embs.shape)
```

```
[load] Loading books from /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/processed/books_with_genres.parquet
[info] books shape: (200000, 10)
[load] Loading authors from /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/data/processed/authors.parquet
[info] authors shape: (829529, 2)
[info] built author_id→name mapping of size 829529
[authors] Resolving author IDs via authors.parquet → author_names...
[authors] Sample author_names:
      book_id      title author_names
0    2767052  The Hunger Games (The Hunger Games, #1)
1         3  Harry Potter and the Sorcerer's Stone (Harry P...
2    41865    Twilight (Twilight, #1)
3     2657    To Kill a Mockingbird
4     4671    The Great Gatsby
5  11870085    The Fault in Our Stars
6     5907    The Hobbit
7     5107    The Catcher in the Rye
8     1885    Pride and Prejudice
9      960    Angels & Demons (Robert Langdon, #1)
[build] Built text corpus of length 200000
[model] Loading text encoder: sentence-transformers/all-MiniLM-L6-v2
[encode] Encoding texts in batches...

100%|          | 1563/1563 [50:50<00:00, 1.95s/it]

[info] desc_embeddings shape: (200000, 384)
[done] Saved description embeddings to /Users/lindseyraven/Desktop/MSAI_SJSU_Work/CMPE259_NatLangProc/Project/Final_Project_Code/CMPE_259_Book_Recommender_FineTune_LLM/models/desc_embeddings.npy
Embeddings shape: (200000, 384)
books_with_genres shape: (200000, 10)
embeddings shape: (200000, 384)
```

1.4 03 - Send Prompts to LLM and Recommender Module

```
[2]: from overarching_module import BookRecSession

     # Instantiate a session (loads DescEmbeddingTeacher under the hood)
```

```

session = BookRecSession()
first_msg = (
    "I really enjoy The Hobbit, can you give me some book recommendations based_
    ↪on that? I would like 5 recommendations. I also do not enjoy the genres_
    ↪young adult, romance. I hate the book Twilight."
)

first_result = session.handle_first_turn(first_msg)

print("=== RAW PREFS FROM PLANNER ===")
import json
print(json.dumps(first_result["preferences"], indent=2))

print("\n=== RECOMMENDATIONS TO SHOW USER ===")
print(first_result["response_text"])

```

[init] Teacher initialized with titles, genres, authors, embeddings (genres=17, authors=66085)
[user] built user embedding from query: 35 liked-book idx (|liked_books|=1), 1 disliked-book idx (|disliked_books|=1), 0 genre-based idx (|liked_genres|=0), 0 author-based idx (|liked_authors|=0), alpha=1.0

===== FILTERED =====

```

764872 |
The Annotated Hobbit |
rating=4.54 |
genres=['fantasy', 'classics'] |
authors=['J.R.R. Tolkien', 'Douglas A. Anderson'] |
score=0.3697
-----
27415414 |
Fool's Gold (The Dragon Lords #1) |
rating=3.85 |
genres=['fantasy', 'science fiction'] |
authors=['Jon Hollins'] |
score=0.3070
-----
4653703 |
Tales from the Perilous Realm |
rating=4.11 |
genres=['fantasy', 'nonfiction', 'classics', 'poetry'] |
authors=['J.R.R. Tolkien', 'Alan Lee'] |
score=0.2993
-----
13636612 |
Mud, Sweat and Tears |
rating=4.15 |
genres=['nonfiction', 'spirituality'] |

```

```

authors=['Bear Grylls'] |
score=0.2982
-----
1804660 |
Journeys of a Lifetime: 500 of the World's Greatest Trips |
rating=4.24 |
genres=['nonfiction', 'spirituality'] |
authors=['National Geographic Society'] |
score=0.2782
-----

===== UNMATCHED =====
liked_books |

disliked_books |

excluded_books |

=== RAW PREFS FROM PLANNER ===
{
  "liked_books": [
    {
      "title": "The Hobbit",
      "rating": 5.0
    }
  ],
  "disliked_books": [
    {
      "title": "Twilight",
      "rating": 1.0
    }
  ],
  "excluded_books": [],
  "liked_genres": [],
  "disliked_genres": [
    "young adult",
    "romance"
  ],
  "liked_authors": [],
  "disliked_authors": [
    "Twilight"
  ],
  "num_recommendations": 5
}

=== RECOMMENDATIONS TO SHOW USER ===
Here are your current book recommendations:

```

```

number=0 |
title: The Annotated Hobbit
average_rating: 4.54
genres: fantasy, classics
authors: J.R.R. Tolkien, Douglas A. Anderson
score: 0.3697
-----
number=1 |
title: Fool's Gold (The Dragon Lords #1)
average_rating: 3.85
genres: fantasy, science fiction
authors: Jon Hollins
score: 0.3070
-----
number=2 |
title: Tales from the Perilous Realm
average_rating: 4.11
genres: fantasy, nonfiction, classics, poetry
authors: J.R.R. Tolkien, Alan Lee
score: 0.2993
-----
number=3 |
title: Mud, Sweat and Tears
average_rating: 4.15
genres: nonfiction, spirituality
authors: Bear Grylls
score: 0.2982
-----
number=4 |
title: Journeys of a Lifetime: 500 of the World's Greatest Trips
average_rating: 4.24
genres: nonfiction, spirituality
authors: National Geographic Society
score: 0.2782
-----

```

```

[3]: feedback_msg = (
    "I dislike book #3, please generate 5 new recommendations."
)

follow_result = session.handle_followup(feedback_msg)

print("=== UPDATED PREFS FROM PLANNER ===")
import json
print(json.dumps(follow_result["preferences"], indent=2))

print("\n=== UPDATED RECOMMENDATIONS TO SHOW USER ===")

```

```
print(follow_result["response_text"])
```

```
=== FINAL FOLLOW-UP PREFS ===
```

```
{
  "liked_books": [
    {
      "title": "The Hobbit",
      "rating": 5.0
    }
  ],
  "disliked_books": [
    {
      "title": "Twilight",
      "rating": 1.0
    }
  ],
  "excluded_books": [
    {
      "title": "Mud, Sweat and Tears"
    }
  ],
  "liked_genres": [],
  "disliked_genres": [
    "young adult",
    "romance"
  ],
  "liked_authors": [],
  "disliked_authors": [
    "Twilight"
  ],
  "num_recommendations": 5
}
```

[user] built user embedding from query: 35 liked-book idx (|liked_books|=1), 1 disliked-book idx (|disliked_books|=1), 0 genre-based idx (|liked_genres|=0), 0 author-based idx (|liked_authors|=0), alpha=1.0
[exclude] 2 indices derived from 1 excluded_books entries.

```
===== FILTERED =====
```

```
764872 |
The Annotated Hobbit |
rating=4.54 |
genres=['fantasy', 'classics'] |
authors=['J.R.R. Tolkien', 'Douglas A. Anderson'] |
score=0.3697
-----
27415414 |
Fool's Gold (The Dragon Lords #1) |
```

```

rating=3.85 |
genres=['fantasy', 'science fiction'] |
authors=['Jon Hollins'] |
score=0.3070
-----
4653703 |
Tales from the Perilous Realm |
rating=4.11 |
genres=['fantasy', 'nonfiction', 'classics', 'poetry'] |
authors=['J.R.R. Tolkien', 'Alan Lee'] |
score=0.2993
-----
1804660 |
Journeys of a Lifetime: 500 of the World's Greatest Trips |
rating=4.24 |
genres=['nonfiction', 'spirituality'] |
authors=['National Geographic Society'] |
score=0.2782
-----
22918271 |
The Rain Wild Chronicles: The Complete 4-Book Collection |
rating=4.46 |
genres=['fantasy', 'science fiction'] |
authors=['Robin Hobb'] |
score=0.2690
-----

===== UNMATCHED =====
liked_books |

disliked_books |

excluded_books |

=== UPDATED PREFS FROM PLANNER ===
{
  "liked_books": [
    {
      "title": "The Hobbit",
      "rating": 5.0
    }
  ],
  "disliked_books": [
    {
      "title": "Twilight",
      "rating": 1.0
    }
  ],

```



```

    "excluded_books": [
      {
        "title": "Mud, Sweat and Tears"
      }
    ],
    "liked_genres": [],
    "disliked_genres": [
      "young adult",
      "romance"
    ],
    "liked_authors": [],
    "disliked_authors": [
      "Twilight"
    ],
    "num_recommendations": 5
  }

```

=== UPDATED RECOMMENDATIONS TO SHOW USER ===
 Here are your current book recommendations:

```

number=0 |
  title: The Annotated Hobbit
  average_rating: 4.54
  genres: fantasy, classics
  authors: J.R.R. Tolkien, Douglas A. Anderson
  score: 0.3697
-----

```

```

number=1 |
  title: Fool's Gold (The Dragon Lords #1)
  average_rating: 3.85
  genres: fantasy, science fiction
  authors: Jon Hollins
  score: 0.3070
-----

```

```

number=2 |
  title: Tales from the Perilous Realm
  average_rating: 4.11
  genres: fantasy, nonfiction, classics, poetry
  authors: J.R.R. Tolkien, Alan Lee
  score: 0.2993
-----

```

```

number=3 |
  title: Journeys of a Lifetime: 500 of the World's Greatest Trips
  average_rating: 4.24
  genres: nonfiction, spirituality
  authors: National Geographic Society
  score: 0.2782
-----

```

```

number=4 |
title: The Rain Wild Chronicles: The Complete 4-Book Collection
average_rating: 4.46
genres: fantasy, science fiction
authors: Robin Hobb
score: 0.2690
-----

```

```

[4]: feedback_msg = (
    "I also am not a fan of the book Journeys of a Lifetime: 500 of the World's
    ↪Greatest Trips, and I dont like the genres of poetry or spirituality."
)

follow_result = session.handle_followup(feedback_msg)

print("=== UPDATED Prefs FROM PLANNER ===")
import json
print(json.dumps(follow_result["preferences"], indent=2))

print("\n=== UPDATED RECOMMENDATIONS TO SHOW USER ===")
print(follow_result["response_text"])

```

```

=== FINAL FOLLOW-UP Prefs ===
{
  "liked_books": [
    {
      "title": "The Hobbit",
      "rating": 5.0
    }
  ],
  "disliked_books": [
    {
      "title": "Twilight",
      "rating": 1.0
    },
    {
      "title": "Journeys of a Lifetime: 500 of the World's Greatest Trips",
      "rating": 1.0
    }
  ],
  "excluded_books": [
    {
      "title": "Mud, Sweat and Tears"
    },
    {
      "title": "Journeys of a Lifetime: 500 of the World's Greatest Trips"
    }
  ],
}

```

```

    "liked_genres": [],
    "disliked_genres": [
        "young adult",
        "romance",
        "poetry",
        "spirituality"
    ],
    "liked_authors": [],
    "disliked_authors": [
        "Twilight"
    ],
    "num_recommendations": 5
}

```

[user] built user embedding from query: 35 liked-book idx (|liked_books|=1), 2 disliked-book idx (|disliked_books|=2), 0 genre-based idx (|liked_genres|=0), 0 author-based idx (|liked_authors|=0), alpha=1.0
[exclude] 3 indices derived from 2 excluded_books entries.

===== FILTERED =====

```

764872 |
The Annotated Hobbit |
rating=4.54 |
genres=['fantasy', 'classics'] |
authors=['J.R.R. Tolkien', 'Douglas A. Anderson'] |
score=0.4709
-----
3141295 |
The Children of Hurin/The Silmarillion/The Hobbit/The Lord of the Ring |
rating=4.72 |
genres=['fantasy', 'science fiction', 'nonfiction', 'classics'] |
authors=['J.R.R. Tolkien'] |
score=0.4132
-----
1066659 |
The Soddit: Or, Let's Cash in Again |
rating=3.07 |
genres=['fantasy', 'science fiction', 'nonfiction', 'classics'] |
authors=['Adam Roberts'] |
score=0.3642
-----
27415414 |
Fool's Gold (The Dragon Lords #1) |
rating=3.85 |
genres=['fantasy', 'science fiction'] |
authors=['Jon Hollins'] |
score=0.3383
-----

```

```

584367 |
Charon: A Dragon at the Gate (The Four Lords of the Diamond, #3) |
rating=3.90 |
genres=['fantasy', 'science fiction', 'mystery', 'nonfiction'] |
authors=['Jack L. Chalker'] |
score=0.3331
-----

===== UNMATCHED =====
liked_books |

disliked_books |

excluded_books |

=== UPDATED PREFS FROM PLANNER ===
{
  "liked_books": [
    {
      "title": "The Hobbit",
      "rating": 5.0
    }
  ],
  "disliked_books": [
    {
      "title": "Twilight",
      "rating": 1.0
    },
    {
      "title": "Journeys of a Lifetime: 500 of the World's Greatest Trips",
      "rating": 1.0
    }
  ],
  "excluded_books": [
    {
      "title": "Mud, Sweat and Tears"
    },
    {
      "title": "Journeys of a Lifetime: 500 of the World's Greatest Trips"
    }
  ],
  "liked_genres": [],
  "disliked_genres": [
    "young adult",
    "romance",
    "poetry",
    "spirituality"
  ],

```

```

    "liked_authors": [],
    "disliked_authors": [
        "Twilight"
    ],
    "num_recommendations": 5
}

```

=== UPDATED RECOMMENDATIONS TO SHOW USER ===
Here are your current book recommendations:

```

number=0 |
  title: The Annotated Hobbit
  average_rating: 4.54
  genres: fantasy, classics
  authors: J.R.R. Tolkien, Douglas A. Anderson
  score: 0.4709
-----

```

```

number=1 |
  title: The Children of Hurin/The Silmarillion/The Hobbit/The Lord of the Rings
  average_rating: 4.72
  genres: fantasy, science fiction, nonfiction, classics
  authors: J.R.R. Tolkien
  score: 0.4132
-----

```

```

number=2 |
  title: The Soddit: Or, Let's Cash in Again
  average_rating: 3.07
  genres: fantasy, science fiction, nonfiction, classics
  authors: Adam Roberts
  score: 0.3642
-----

```

```

number=3 |
  title: Fool's Gold (The Dragon Lords #1)
  average_rating: 3.85
  genres: fantasy, science fiction
  authors: Jon Hollins
  score: 0.3383
-----

```

```

number=4 |
  title: Charon: A Dragon at the Gate (The Four Lords of the Diamond, #3)
  average_rating: 3.90
  genres: fantasy, science fiction, mystery, nonfiction
  authors: Jack L. Chalker
  score: 0.3331
-----

```

1.5 04 - Final Recommender Bot Example

```
[10]: from overarching_module import BookRecSession

session = BookRecSession()

# First turn
user_msg = input("Describe your preferences (first turn):\n> ")
first_result = session.handle_first_turn(user_msg)
print("\n=== RECOMMENDATIONS ===")
print(first_result["response_text"])

while True:
    more = input("\nGive feedback about these recs (or type 'quit'):\n> ")
    if more.strip().lower() in {"quit", "exit"}:
        print("Ending session.")
        break

    follow_result = session.handle_followup(more)
    print("\n=== UPDATED RECOMMENDATIONS ===")
    print(follow_result["response_text"])
```

[init] Teacher initialized with titles, genres, authors, embeddings (genres=17, authors=66085)

[user] liked_genres contributed from 55184 book embeddings.

[user] built user embedding from query: 535 liked-book idx (|liked_books|=1), 0 disliked-book idx (|disliked_books|=0), 55184 genre-based idx (|liked_genres|=1), 0 author-based idx (|liked_authors|=0), alpha=1.0

===== FILTERED =====

```
1053996 |
Rediscovery (Darkover - Chronological Order #11) |
rating=3.82 |
genres=['fantasy', 'science fiction', 'nonfiction'] |
authors=['Marion Zimmer Bradley', 'Mercedes Lackey'] |
score=0.7152
-----
33384094 |
Dangerous Ways (The Books of Winter, #1) |
rating=4.69 |
genres=['fantasy', 'science fiction'] |
authors=['R.R. Virdi'] |
score=0.7015
-----
24500161 |
Blood Red (Mundy's Landing, #1) |
rating=3.74 |
genres=['mystery', 'horror'] |
```

```

authors=['Wendy Corsi Staub'] |
score=0.6998
-----

===== UNMATCHED =====
liked_books |

disliked_books |

excluded_books |

=== RECOMMENDATIONS ===
Here are your current book recommendations:

number=0 |
title: Rediscovery (Darkover - Chronological Order #11)
average_rating: 3.82
genres: fantasy, science fiction, nonfiction
authors: Marion Zimmer Bradley, Mercedes Lackey
score: 0.7152
-----
number=1 |
title: Dangerous Ways (The Books of Winter, #1)
average_rating: 4.69
genres: fantasy, science fiction
authors: R.R. Virdi
score: 0.7015
-----
number=2 |
title: Blood Red (Mundy's Landing, #1)
average_rating: 3.74
genres: mystery, horror
authors: Wendy Corsi Staub
score: 0.6998
-----
=== FINAL FOLLOW-UP PREFS ===
{
  "liked_books": [
    {
      "title": "Red Rising",
      "rating": 5.0
    }
  ],
  "disliked_books": [],
  "excluded_books": [
    {
      "title": "Dangerous Ways (The Books of Winter, #1)"

```

```

    }
  ],
  "liked_genres": [
    "science fiction"
  ],
  "disliked_genres": [
    "romance",
    "young adult"
  ],
  "liked_authors": [
    "Darkover"
  ],
  "disliked_authors": [],
  "num_recommendations": 4
}

```

[user] liked_genres contributed from 55184 book embeddings.
 [user] built user embedding from query: 535 liked-book idx (|liked_books|=1), 0
 disliked-book idx (|disliked_books|=0), 55184 genre-based idx
 (|liked_genres|=1), 0 author-based idx (|liked_authors|=1), alpha=1.0
 [exclude] 1 indices derived from 1 excluded_books entries.

===== FILTERED =====

```

1053996 |
Rediscovery (Darkover - Chronological Order #11) |
rating=3.82 |
genres=['fantasy', 'science fiction', 'nonfiction'] |
authors=['Marion Zimmer Bradley', 'Mercedes Lackey'] |
score=0.7152
-----
24500161 |
Blood Red (Mundy's Landing, #1) |
rating=3.74 |
genres=['mystery', 'horror'] |
authors=['Wendy Corsi Staub'] |
score=0.6998
-----
13575986 |
Rapture (Bel Dame Apocrypha #3) |
rating=4.00 |
genres=['fantasy', 'science fiction', 'nonfiction', 'spirituality'] |
authors=['Kameron Hurley'] |
score=0.6943
-----
25752594 |
Shivaree |
rating=3.68 |
genres=['fantasy', 'science fiction', 'mystery', 'historical', 'horror'] |

```



```

authors=['J.D. Horn'] |
score=0.6924
-----

===== UNMATCHED =====
liked_books |

disliked_books |

excluded_books |

=== UPDATED RECOMMENDATIONS ===
Here are your current book recommendations:

number=0 |
title: Rediscovery (Darkover - Chronological Order #11)
average_rating: 3.82
genres: fantasy, science fiction, nonfiction
authors: Marion Zimmer Bradley, Mercedes Lackey
score: 0.7152
-----
number=1 |
title: Blood Red (Mundy's Landing, #1)
average_rating: 3.74
genres: mystery, horror
authors: Wendy Corsi Staub
score: 0.6998
-----
number=2 |
title: Rapture (Bel Dame Apocrypha #3)
average_rating: 4.00
genres: fantasy, science fiction, nonfiction, spirituality
authors: Kameron Hurley
score: 0.6943
-----
number=3 |
title: Shivaree
average_rating: 3.68
genres: fantasy, science fiction, mystery, historical, horror
authors: J.D. Horn
score: 0.6924
-----

=== FINAL FOLLOW-UP PREFS ===
{
  "liked_books": [
    {
      "title": "Red Rising",

```

```

        "rating": 5.0
    }
],
"disliked_books": [],
"excluded_books": [
    {
        "title": "Dangerous Ways (The Books of Winter, #1)"
    }
],
"liked_genres": [
    "science fiction"
],
"disliked_genres": [
    "romance",
    "young adult"
],
"liked_authors": [
    "Darkover"
],
"disliked_authors": [],
"num_recommendations": 5
}

```

[user] liked_genres contributed from 55184 book embeddings.
[user] built user embedding from query: 535 liked-book idx (|liked_books|=1), 0
disliked-book idx (|disliked_books|=0), 55184 genre-based idx
(|liked_genres|=1), 0 author-based idx (|liked_authors|=1), alpha=1.0
[exclude] 1 indices derived from 1 excluded_books entries.

===== FILTERED =====

```

1053996 |
Rediscovery (Darkover - Chronological Order #11) |
rating=3.82 |
genres=['fantasy', 'science fiction', 'nonfiction'] |
authors=['Marion Zimmer Bradley', 'Mercedes Lackey'] |
score=0.7152
-----
24500161 |
Blood Red (Mundy's Landing, #1) |
rating=3.74 |
genres=['mystery', 'horror'] |
authors=['Wendy Corsi Staub'] |
score=0.6998
-----
13575986 |
Rapture (Bel Dame Apocrypha #3) |
rating=4.00 |
genres=['fantasy', 'science fiction', 'nonfiction', 'spirituality'] |

```

```

authors=['Kameron Hurley'] |
score=0.6943
-----
25752594 |
Shivaree |
rating=3.68 |
genres=['fantasy', 'science fiction', 'mystery', 'historical', 'horror'] |
authors=['J.D. Horn'] |
score=0.6924
-----

```

===== UNMATCHED =====

liked_books |

disliked_books |

excluded_books |

=== UPDATED RECOMMENDATIONS ===

Here are your current book recommendations:

```

number=0 |
title: Rediscovery (Darkover - Chronological Order #11)
average_rating: 3.82
genres: fantasy, science fiction, nonfiction
authors: Marion Zimmer Bradley, Mercedes Lackey
score: 0.7152
-----

```

```

number=1 |
title: Blood Red (Mundy's Landing, #1)
average_rating: 3.74
genres: mystery, horror
authors: Wendy Corsi Staub
score: 0.6998
-----

```

```

number=2 |
title: Rapture (Bel Dame Apocrypha #3)
average_rating: 4.00
genres: fantasy, science fiction, nonfiction, spirituality
authors: Kameron Hurley
score: 0.6943
-----

```

```

number=3 |
title: Shivaree
average_rating: 3.68
genres: fantasy, science fiction, mystery, historical, horror
authors: J.D. Horn

```

```
score: 0.6924
-----
Ending session.
```